

System of Homogeneous and Non- Homogeneous equations

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Homogeneous linear system:

- ❖ A system of linear equations is said to be homogeneous if the constant terms are all zeroes; that is the system has the form:

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = 0$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = 0$$

.....

.....

$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = 0$$

$$AX=0$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{1n} \\ a_{21} & a_{22} & a_{2n} \\ a_{m1} & a_{m2} & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Augmented Matrix=[A]

- ❖ Every homogeneous system of linear equation is consistent because all such system have $x_1 = 0, x_2 = 0, \dots, x_n = 0$ as a solution.
- ❖ This solution is called the trivial solution; if there are other solution, they are called non-trivial solutions.

Non-Homogeneous linear system:

- Consider the set of 'm' linear equations in 'n' unknowns

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$

.....

.....

$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_n$$

- The system of equations can be written in the matrix notation form,

$$AX=B$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{1n} \\ a_{21} & a_{22} & a_{2n} \\ a_{m1} & a_{m2} & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_n \end{bmatrix}$$

$$\text{Augmented Matrix} = [A \quad B]$$

- ▶ If $\rho(A) = \rho(A \ B) = \text{no. of variables}(n)$ then given system of non – homogeneous equations are consistent and it has unique solution.
- ▶ If $\rho(A) = \rho(A \ B) < \text{no. of variables}(n)$ then given system of non – homogeneous equations are consistent but it has infinite number of solutions. r variables are linearly independent.
- ▶ If $\rho(A) \neq \rho(A \ B)$ then given system of non – homogeneous equations are inconsistent and it has no soluti

Example of Homogeneous linear system:

1. Solve the system of equations $x+3y-2z=0$, $2x-y+4z=0$, $x-11y+14z=0$

Sol: We write the given system in $AX=0$ i.e.,

$$\begin{bmatrix} 1 & 3 & -2 \\ 2 & -1 & -4 \\ 1 & -11 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$R_3 \rightarrow R_3 - R_1$$

$$A = \begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 8 \\ 0 & -14 & 16 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_2$$

$$A = \begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 8 \\ 0 & 0 & 0 \end{bmatrix}$$

The rank of the $A=2$ i.e. $\rho(A)=2 < \text{number of unknowns}=3$

We have infinite no. of solutions

Above matrix we can write as $x+3y-2z=0$, $-7y+8z=0$, $0=0$

Let $z=k$ then $y=8k/7$ & $x=-10k/7$

Giving different values to k , we get infinite no. of values of x, y, z .

Example of Non-Homogeneous linear system:

1. Solve the system of equations $x+y+z=6$; $x-y+2z=5$; $3x+y+z=-8$

Sol: We write the given system in $AX=B$ i.e.,

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \\ 3 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 5 \\ -8 \end{bmatrix}$$

Augmented matrix = $[A \ B]$

$$[A \ B] = \begin{bmatrix} 1 & 1 & 1 & 6 \\ 1 & -1 & 2 & 5 \\ 3 & 1 & 1 & -8 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 - 3R_1$$

$$[A \ B] = \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & -2 & 1 & -1 \\ 0 & -2 & -2 & -26 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$[A \ B] = \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & -2 & 1 & -1 \\ 0 & 0 & -3 & -25 \end{bmatrix}$$

Here, it is in echelon form

$$\rho(A) = \rho(A \ B) = 3$$

The given system of equations are consistent

No. of unknowns $n=3$

$$\rho(A) = \rho(A \ B) = n$$

Here we get unique solution

$$\begin{array}{rcl} \text{From the above matrix } x+y+z & = & 6 \text{-----} \textcircled{1} \\ -2y+z & = & -1 \text{-----} \textcircled{2} \\ -3z & = & -25 \text{-----} \textcircled{3} \end{array}$$

From $\textcircled{3}$

$$-3z = -25$$

$$Z = 25/3$$

From $\textcircled{2}$

$$-2y = -1 - 25/3 \rightarrow -2y = -28/3 \rightarrow y = 14/3$$

From $\textcircled{1}$

$$x + 14/3 + 25/3 = 6 \rightarrow x + 39/3 = 6 \rightarrow x + 13 = 6 \rightarrow x = -7$$



Thank you